26

# Core Principles of Perioperative Management in Children

*Sherry S. Ross, and Janey R. Phelps*

## Keywords

comorbidities; pediatric anesthesia; pediatric pain management; surgical complications

## Questions

1. Current U.S. Food and Drug Administration (FDA) guidelines for the administration of anesthesia agents to children are as follows:
    a. Repeated or lengthy use of general anesthetic and sedation drugs during surgeries or procedures in children younger than 3 years or in pregnant women during their third trimester may affect the development of children's brains.
    b. Repeated or lengthy use of general anesthetic and sedation drugs during surgeries or procedures in children younger than 2 years or in pregnant women during their third trimester may affect the development of children's brains.
    c. Repeated or lengthy use of general anesthetic and sedation drugs during surgeries or procedures in children younger than 3 years or in pregnant women during their second trimester may affect the development of children's brains.
    d. Repeated or lengthy use of general anesthetic and sedation drugs during surgeries or procedures in children younger than

3 years may affect the development of children's brains.

e. Repeated or lengthy use of general anesthetic and sedation drugs during surgeries or procedures in children younger than 2 years or in pregnant women during their second trimester may affect the development of children's brains.

2. Which is true regarding pain in children?
    a. Discussion of pain expectations with children and parents influences pain associated with procedures.
    b. Child life specialists are experts in the education and procedure preparation of children and families through various techniques that help overcome anxiety and should be used in the perioperative setting to help decrease child and parental anxiety.
    c. Preoperative anxiety in children is predictive of postoperative pain levels and behaviors.
    d. a and c
    e. a, b, and c
3. Which of the following concerning pain assessment in children is FALSE?
    a. Self-assessment of pain is considered the "gold standard" for pain assessment.
    b. Self-assessment measurements to rate pain are not present in children less than 2 years old.
    c. Self-assessment in children include: validated in children include Pieces of Hurt (3–6 years), Faces scale (4–12 years), Oucher scale (4–12 years), visual analog scale (8–16 years).
    d. Objective pain scales are available for premature infants (e.g., Premature Infant Pain Profile [PIPP]), neonates, and nonverbal and cognitively impaired children.
    e. Pain should be described in terms of intensity, location, duration, modifiable interventions, and quality.
4. NPO (nothing by mouth [nil per os]) status recommendations in children include:
    a. There should be an 8-hour period between the intake of full meals prior induction of general anesthesia, regional anesthesia, or sedation/analgesia.

b. Children may consume formula, nonhuman milk, or a light meal up to 6 hours prior to needed anesthesia.
c. There should be a 4-hour period of time between the consumptions of breast milk and administration of general anesthesia, regional anesthesia, or sedation/analgesia.
d. Children may have clear liquids up to 2 hours before elective procedures requiring anesthesia administration since risk of aspiration is low.
e. All of the above.

5. Children may have dietary requirements outside the normal parameters. Which of the following is a false statement?
   a. Children on a ketogenic diet should receive carbohydrate-free fluids.
   b. Enteral feeds are considered a fatty meal and should be stopped 8 hours prior to anesthesia induction.
   c. Children with diabetes may be affected by volume depletion during preoperative fasting coupled with the discontinuation of outpatient diabetes medications. This may result in preoperative hypoglycemia.
   d. Metformin should be discontinued 24 hours prior to surgery because it has a long half-life, which increases the risk of lactic acidosis in the presence of dehydration, hypoxemia, or poor tissue perfusion.
   e. Medications that should be taken on the morning of surgery include antiseizure, gastrointestinal reflux, and asthma medications.
6. Which of the following is true concerning perioperative laboratory evaluation?
   a. Urologic procedures where the urinary tract is opened electively are considered clean contaminated and increase the risk of postoperative infection.
   b. Routine laboratory evaluation has little use and is not cost effective.
   c. For female patients of childbearing age, a preoperative urine pregnancy test is recommended.

d. The American Academy of Pediatrics recommended preoperative chest x-ray as part of routine preoperative assessment.
e. a, b, and c.

7. Perioperative considerations in premature infants include:
    a. Respiratory function in the premature infant is challenging due to inadequate surfactant levels, especially in children born less than 36 weeks, resulting in high oxygen consumptions and low pulmonary functional residual capacity, as well as tracheomalacia.
    b. Premature infants have a high risk of apnea, which can result in death.
    c. Premature infants have an increased risk of retinopathy and blindness in the presence of supplemental oxygen so the overall goal is oxygen saturation percentage ranging in the low 90s.
    d. Neonates have a higher potential for large fluid losses due to higher rates of evaporation from skin and surgical wound sights, as well as decreased ability to concentrate urine.
    e. b, c, and d.
8. Regarding children with renal insufficiency, which of the following is true?
    a. Infants and children under the age of 6 years typically have kidney insufficiency due to acquired abnormalities such as immunoglobulin A (IgA) nephropathy.
    b. Children with chronic kidney disease (CKD) have an increased potential for anemia due to decreases in the renal production of erythropoietin and increases in ureas, which decreases red blood cell production.
    c. Elevations in urea platelet dysfunction, which increases the risk of thrombus.
    d. Cardiac function is typically normal in patients with CKD.
    e. Electrolyte abnormalities in children with CKD are often abnormal. Abnormalities such as hypernatremia can increase the risk of cardiac arrhythmias.
9. Which of the follow is a true statement?

a. Patients with CKD 2 have mild impairment of renal function but are able to excrete drugs and metabolites, so they do not require medication adjustments. However, they should be monitored closely for effects of NSAIDs on renal function and for worsening renal deterioration.
b. The most important factor in the perioperative management of children with sickle cell disease is dehydration, especially during the preoperative NPO status.
c. In the presence of high-inspired oxygen levels during anesthesia, patients treated with the chemotherapeutic agent bleomycin can develop pulmonary toxicity, which can be fatal.
d. b and c only.
e. All of the above.

10. Which of the following statements are true concerning children under anesthesia for surgical procedures?
    a. Spina bifida patients with lesions at T5 to T8 are at an increased risk of autonomic dysreflexia with noxious stimuli. This can lead to uncontrolled hypertension, bradycardia, stroke, and potentially death.
    b. Spina bifida patients have an increased risk of latex allergies.
    c. Symptoms of severe latex allergy include bronchoconstriction, vasodilation and subsequent hypotension, increased capillary permeability, coronary vasoconstriction, and myocardial depression.
    d. Children with congenital adrenal hyperplasia can have epinephrine and cortisol deficiency, which may result in hypoglycemia.
    e. All of the above.
11. Which of the following is a false statement?
    a. For postoperative pain management in the septic child, peripheral nerve blocks may be effective and avoid the systemic effects of opioid in a hemodynamically unstable patient.
    b. Pediatric trauma victims should always be considered to have a full stomach, and rapid sequence induction should be the gold standard.

c. Infection-induced inflammation of the upper respiratory mucosa will result in airway hyperreactivity, which increases secretions and airway irritability during anesthesia and can persist for 2 to 8 weeks following infection.
d. The American Urological Association guidelines state that if urologic surgery is planned in the presence of an existing infection, antibiotics should be administered with the goal of a sterilized operative field.
e. All of the above.

12. Which of the following are true?
    a. The most common cause of perioperative fever in children involves an infectious etiology, which is most commonly a viral upper respiratory tract infection.
    b. Preoperative urine cultures should be obtained preoperatively if a sterilized field is required and can be obtained.
    c. Blood transfusion can result in a nonspecific fever up to 1 week after transfusion.
    d. Preoperative UTI, especially if recurrent, is recognized as a high-risk factor for postoperative infection.
    e. All of the above.
13. Which of the following statements are true?
    a. The ERAS (Early Recovery After Surgery) pathway uses several components, which include avoiding prolonged fasting, early enteral nutrition, preoperative oral carbohydrate loading, early mobilization, and enhanced efforts in patient education. These are yet to be established in pediatric patients.
    b. In the preoperative setting, administration of antibiotics should allow time for adequate serum and tissue concentrations prior to incision.
    c. Neuromuscular blocking agents and β-lactam antibiotics are the main triggers of perioperative anaphylactic reactions.
    d. Treatment of perioperative anaphylaxis includes stopping the suspected drug immediately, 100% oxygen administration, Trendelenburg position, fluid therapy, and epinephrine.
    e. All of the above.
14. Which of the following is true concerning malignant hyperthermia?

a. Malignant hyperthermia is a common disorder
b. Malignant hyperthermia is a potentially fatal hypermetabolic disorder that is triggered by acetylcholine and inhaled anesthetics.
c. Signs of malignant hyperthermia include hypocarbia, sinus bradycardia, masseter muscle spasm, and hyperthermia.
d. If malignant hyperthermia is suspected, the triggering agent should be stopped, anesthesia backup should be requested, anesthesia machine should be exchanged, and an arterial line should be placed, as well as a Foley catheter.
e. Late signs include severe metabolic alkylosis, hypokalemia, generalized muscle rigidity, and rhabdomyolysis.

15. Which of the following is true concerning the postoperative period?
    a. Respiratory events only include hypoxia and airway obstruction.
    b. Most postoperative cardiac events in children are preventable.
    c. Emergency delirium occurs only in children between the ages of 2 to 8.
    d. Patients with high preoperative anxiety are at increased risk of postoperative emergence delirium.
    e. b and d.

## Answers

1. **a. Repeated or lengthy use of general anesthetic and sedation drugs during surgeries or procedures in children younger than 3 years or in pregnant women during their third trimester may affect the development of children's brains.**
2. **e. a, b, and c.** Discussion of pain expectations with children and parents influences pain associated with procedures. **Child life specialists are experts in the education and procedure preparation of children and families** through various techniques that help to overcome anxiety and should be used in the perioperative setting to help decrease child and parental anxiety.

Preoperative anxiety in children is predictive of postoperative pain levels and behaviors.

3. **b. Self-assessment measurements to rate pain are not present in children less than 2 years old. "Self-assessment measurements to rate pain are not present in children less than 2 years old" is a false statement.**
4. **e. All of the above.** NPO status recommendations in children include all of the listed options.
5. **c. Children with diabetes may be affected by volume depletion during preoperative fasting coupled with the discontinuation of outpatient diabetes medications. This may result in preoperative hypoglycemia.** "Children with diabetes may be affected by volume depletion during preoperative fasting coupled with the discontinuation of outpatient diabetes medications. This may result in preoperative hypoglycemia" is a false statement.
6. **e. a, b, and c.** All of the listed options are true.
7. **e. b, c, and d. Premature infants have a high risk of apnea,** which can result in death. Premature infants have an increased risk of retinopathy and blindness in the presence of supplemental oxygen, so the overall goal is oxygen saturation percentage ranging in the low 90s. **Neonates have a higher potential for large fluid losses** due to higher rates of evaporation from skin and surgical wound sites, as well as decreased ability to concentrate urine.
8. **b. Children with chronic kidney disease (CKD) have an increased potential for anemia due to decreases in the renal production of erythropoietin and increases in urea, which decreases red blood cell production.**
9. **e. All of the above.** All of the listed options are true.
10. **e. All of the above.** All of the listed options are true.
11. **e. All of the above.** All of the listed options are true.
12. **e. All of the above.** All of the listed options are true.
13. **e. All of the above.** All of the listed options are true.
14. **d.** If malignant hyperthermia is suspected, the triggering agent should be stopped, anesthesia backup should be requested, anesthesia machine should be exchanged, and an arterial line should be placed, as well as a Foley catheter.

15. **e. b, and d.** Most postoperative cardiac events in children are preventable. **Patients with high preoperative anxiety are at increased risk of postoperative emergence delirium.**

## Chapter Review

1. Families should be informed of current FDA recommendations including what is known and what studies are currently underway to allow thoroughly informed consent.
2. Anesthesia risks include nausea, vomiting, bronchospasm, laryngospasm, stridor, emergence delirium, and postanesthesia maladaptive behaviors.
3. NPO violation significantly increases risk of gastric aspiration and pneumonitis.
4. Premature infants have a significantly increased risk of perioperative complications.
5. Spina bifida patients have the potential for multiple comorbidities that should be considered perioperatively.
6. Autonomic dysreflexia is life threatening, and the pediatric urologist should know signs and symptoms of onset.
7. Antibiotics are the mainstay of sepsis treatment and should be administered as soon as possible. Cultures should be obtained if possible.
8. Urine cultures should be obtained preoperatively when indicated and positive cultures treated with antibiotics based on sensitivities.
9. Antibiotics are the mainstay to prevent surgical site infections. Selection of antimicrobial should consider the site and potential source of infection.
10. Pain management is important to discuss with anesthesia for intraoperative and postoperative management.
11. Perioperative complications most often occur in the immediate postoperative period.
12. Postoperative pain control is important for optimal surgical outcomes. A multimodal drug approach is recommended.

13. Proper administration and disposal of unused narcotics are necessary to prevent overdoses with these medications.

27

# Principles of Laparoscopic and Robotic Surgery in Children

*Thomas Sean Lendvay, and Jonathan Ellison*

## Keywords

laparoscopy; minimally invasive surgery; pediatric urology

## Questions

1. Which of the following is *true* regarding cosmetic benefits of minimally invasive surgery in children?
    a. Incisional scars will contract over time as the children grow longitudinally.
    b. Parents have no preference toward clinical outcomes (success and complications) over cosmetic appears of incisions when weighing open versus minimally invasive approach.
    c. Size of the incisions are not as important as location of the incision as it relates to cosmetic outcomes.
    d. Cosmetic benefits of single-port laparoendoscopic surgery are durable over several years.
    e. No validated questionnaires exist to assess bother as related to incisional scarring following surgical interventions.
2. Which of the following procedures has not been shown to have a length of stay benefit for minimally invasive approaches as compared with the open surgical counterpart?
    a. Pyeloplasty

b. Ureteral reimplant
c. Hernia/hydrocele
d. Appendicovesicostomy
e. Augmentation enterocystoplasty

3. Which of the following is false regarding cardiovascular implications of pneumoperitoneum?
   a. Bradycardia may be seen upon insufflation.
   b. Tachyarrhythmia may be seen secondary to a sympathetic response of pneumoperitoneum.
   c. Cardiac index is decreased at insufflation pressures as low as 10 mm Hg.
   d. Left ventricular ejection fraction does not appear to be impacted by pneumoperitoneal changes, although myocardial workload may increase.
   e. Mortality is no different in laparoscopic approaches for children with congenital cardiac defects as compared with healthy children, despite the cardiovascular implications of pneumoperitoneum.
4. With pneumoperitoneum, which of the following physiological changes is incorrectly described?
   a. Increase in end-tidal $CO_2$
   b. Increase in dynamic pulmonary compliance
   c. Increase in systemic vascular resistance
   d. Increase in intracranial pressure
   e. Decrease in urine output
5. Which of the following statements appropriately reflects the impact of the pneumoperitoneum on the immune response?
   a. Polymorphonuclear cells and macrophages, as well as levels of interleukin-6 (IL-6), are increased locally due to the effect of pneumoperitoneum across the entire abdominal wall, as compared with open surgery.
   b. Pediatric patients have a greater degree of benefit toward a blunted immune response following laparoscopy as compared with adult patients.

c. Peritoneal signs following laparoscopy are heightened as compared with open surgery, likely owing to the local acidotic environment following $CO_2$ insufflation.

d. Systemic C-reactive protein (CRP) and IL-6 levels are decreased following laparoscopy as compared with open surgery.

e. Insufflation with room air results in a lessened inflammatory response as compared with $CO_2$ gas.

6. Which of the following is/are an absolute contraindication to minimally invasive surgery?
   a. Size less than 4 kg
   b. Inability to tolerate pneumoperitoneum
   c. Age less than 3 months
   d. Prior abdominal surgery
   e. a and c
7. Which of the following is true regarding peritoneal access via the umbilicus?
   a. A fascial hernia is rarely identified.
   b. Due to the fascial support at the umbilicus, a port site in this location does not usually require closure.
   c. The umbilicus is a reliable indicator of the location of the aortic bifurcation, which will lie superior to the umbilicus.
   d. Trendelenburg positioning does not alter the relation between the umbilicus and the aortic bifurcation.
   e. None of the above.
8. A comparison of open and Veress access techniques has revealed:
   a. an increased risk of visceral injury with open access.
   b. an increased risk of vascular injury with Veress access.
   c. an increased risk of failed access with Veress access.
   d. an increased risk of failed access with open access.
   e. an increased risk of port site hernia with open access.
9. Port site hernias in children:
   a. have not been reported in trocar sites smaller than 5 mm.
   b. may present with lethargy and a bulge in the absence of peritoneal signs.

c. are at an increased risk at the umbilical site.
d. are a lower risk complication after robotic-assisted procedures.
e. b and c.

10. The blood vessel most likely injured by Veress needle access at the umbilicus, as shown in Fig. 27.1, is:
    a. left common iliac vein.
    b. right common iliac vein.
    c. right common iliac artery.
    d. vena cava.
    e. aorta.
11. Pulmonary air embolus:
    a. may be prevented by using $CO_2$ insufflation.
    b. is a common complication with rare catastrophic outcome.

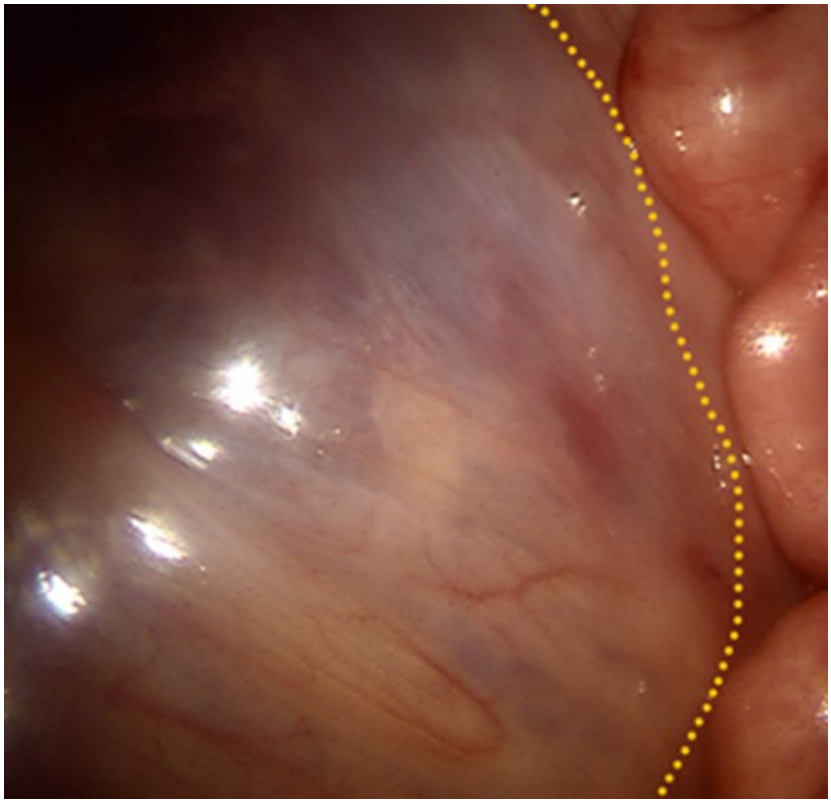


FIG. 27.1

c. may present with increased or decreased end-tidal $CO_2$.

d. likely results from direct absorption of insufflation gas via the peritoneum.
e. can be managed best by repositioning the patient prone.

12. All of the following factors are aid in training of minimally invasive surgeons EXCEPT:
    a. training standardization across centers.
    b. standardized assessments of robotic and laparoscopic surgical skills.
    c. certified curriculum for surgical training.
    d. well-validated low- and high-fidelity simulation.
    e. use of surgical "warm-up."

## Answers

1. **c. Size of the incisions are not as important as location of the incision as it relates to cosmetic outcomes.** Validated instruments to assess scar both do exist and have been used to show that families show preference over location of scar (i.e., prefer more inconspicuous positions), as well as that the benefits cosmetically from a single-site incisional approach are not significantly better than standard laparoscopic approaches after 18 months of follow-up. Families would prefer optimal clinical outcomes to cosmetic appearance, although if equal, families would choose a minimally invasive approach for their child.
2. **c. Hernia/hydrocele.** Although a combination of prospective observational trials, randomized controlled trials, and retrospective cohort analyses have shown length of stay benefits for minimally invasive pyeloplasty, ureteral reimplantation, appendicovesicostomy, and augmentation enterocystoplasty, no such data exist showing benefit for hernia/hydrocele repairs. In fact, pain scores may be higher following laparoscopic hernia/hydrocele repairs as compared with open inguinal incisions, potentially due to the stress of pneumoperitoneum. Thus the benefit of convalescence and postoperative pain from a minimally invasive

approach is likely proportional to the burden of the alternative open approach.

3. **e. Mortality is no different in laparoscopic approaches for children with congenital cardiac defects as compared with healthy children, despite the cardiovascular implications of pneumoperitoneum.** Vagal nerve stimulation can result in initial bradycardia, which may be managed with atropine or glycopyrrolate but is typically self-limited. Cardiac index decreases with pneumoperitoneum greater than 10 mm Hg, likely due in part to increased intraabdominal and intrathoracic pressure on the venous return. This may result in an increased work of the myocardium as a compensatory response, which is why the left ventricular ejection fraction remains stable in spite of the cardiovascular changes. However, mortality is increased for children with congenital cardiac defects undergoing laparoscopic procedures as compared with healthy children, indicating a need for close hemodynamic monitoring and consideration of a specialized cardiac anesthesia team for these procedures.
4. **b. Increase in dynamic pulmonary compliance.** End-tidal $CO_2$ will increase from both an increased absorption of insufflated $CO_2$ gas and a decrease in minute ventilation from upward pressure on the diaphragm. Systemic vascular resistance increases, which drives a sympathetic response. Both of these changes likely contribute to a decrease in glomerular filtration rate and thus a decrease in urine output. Intracranial pressure also increases, although in most cases not to a clinically relevant degree. Dynamic pulmonary compliance will decrease, likely secondary to upward mobilization of the diaphragm and a decrease in the overall tidal volumes; often these changes necessitate ventilator manipulations to maintain appropriate ventilation.
5. **d. Systemic C-reactive protein (CRP) and IL-6 levels are decreased following laparoscopy as compared with open surgery.** Systemic and local inflammatory responses are blunted with insufflation, a result that appears to be in part due to the use of $CO_2$ gas, which elicits a lessened immune response as compared

with insufflation with room air or open surgery. This blunted immune response results in lower pain and fatigue levels postoperatively but also may blunt peritoneal signs in the event of a visceral injury. Thus the surgeon must have a high index of suspicion for visceral injury in the absence of an acute abdomen, if the child otherwise appears ill.

6. **b. Inability to tolerate pneumoperitoneum.** Inability to tolerate pneumoperitoneum and inability of the surgeon to perform the procedure in a minimally invasive fashion are the only two absolute contraindications to a minimally invasive approach in children. Children less than a week of age have safely undergone robotic and laparoscopic-assisted procedures in experienced hands. Although prior abdominal operations may increase the risk for conversion to open surgery, these patients may also experience a differential benefit to a minimally invasive approach if performed successfully.
7. **e. None of the above.** Fascial openings are commonly observed in children and even found in up to 25% of adults at the umbilicus. However, the variance of the fascial support necessitates closure of the umbilical port sites in children. The umbilicus is a poor indicator of the location of the aortic bifurcation, which may fall inferior to the umbilicus in up to 40% of patients, a proportion that may even increase should the patient be placed in Trendelenburg position. The implication of this fact is that the surgeon should take care in blind trocar placement at the umbilicus as major vascular structures including the aorta and vena cava are at risk of injury in that approach.
8. **c. An increased risk of failed access with Veress access.** Major complications with access are rare regardless of technique chosen. Thus no difference is observed in major vascular or visceral injury with either open or Veress access technique. However, Veress access does appear to be associated with a higher risk for access failure as compared with the open technique. Port site hernia risk has not been formally assessed in most studies.
9. **b. May present with lethargy and a bulge in the absence of peritoneal signs.** Port site herniation of omentum has been reported in a 3-mm port site, indicating the importance of closure of all

laparoscopic and robotic ports in children. Unlike adults, there does not appear to be a protective location for port site hernia comparing midline, umbilical, and lateral port site location. The blunted inflammatory response post-pneumoperitoneum can mute peritoneal signs even when a port site herniation poses risk of compromise to incarcerated intestine.

10. **a. Left common iliac vein.** This photo represents a retroperitoneal hematoma from venous injury, the dotted line representing the extent of the hematoma bulge. Although the umbilicus is a poor marker of the aortic bifurcation, most commonly the left common iliac vein is found as the most superior blood vessel in this location, crossing over the iliac artery. Recognition of a vascular injury at the time of Veress access is imperative as dilation with a trocar in this situation could result in life-threatening hemorrhage.
11. **c. May present with increased or decreased end-tidal ($CO_2$ )** Although traditionally described as presenting with an increase in end-tidal $CO_2$, vascular collapse with either increased or decreased end-tidal $CO_2$ at the onset of insufflation should prompt concern for a pulmonary air embolus. This rare but life-threatening event may be managed by placing the patient in the left lateral decubitus position, thereby encouraging migration of the air embolus out of the right pulmonary vein back into the right atrium. Up to 10% of insufflated gas initially may be nitrogen or oxygen, even when using $CO_2$ insufflation, emanating from the insufflation tubing. Thus flushing the tubing is recommended, especially in infants where the volume of ventilator dead space might be quite low.
12. **a. Training standardization across centers.** Although a certified training curriculum (Fundamentals in Laparoscopic Surgery) exists and is commonly applied across general surgery training sites, there is not a standardized laparoscopic or robotic curriculum for urologic surgeons. However, standardized assessments, a combination of low- and high-fidelity surgical simulators, and surgical warm-ups are all tools that can aid in training the newest generation of minimally invasive urologic surgeons, as well as maintaining expertise and skill in those already practicing.

## Chapter Review

1. A diminished postoperative inflammatory response may result in postoperative complications such as visceral injury presenting without peritoneal signs.
2. Because of the risk of herniation, closure of all port sites in the pediatric population following minimally invasive surgery is recommended.
3. Infants pose anatomic and physiologic challenges such as limited working space, a pliable abdominal wall, and cardiopulmonary changes with insufflation.
4. Benefits of smaller laparoscopic instrumentation must be weighed against potential limited versatility of the smallest instruments currently on the market.
5. Novel instrumentation, robotic technology, and augmented intelligence will continue to further the efficiency and safety of minimally invasive surgery.
6. Surgeon simulation, standardized evaluation and feedback, and warm-up preparation before surgery may all play a role in enhancing a surgeon's preparedness for minimally invasive surgery.

SECTION

# Lower Urinary Tract Conditions

## OUTLINE


28. Clinical and Urodynamic Evaluation of Lower Urinary Tract Dysfunction in Children

29. Management Strategies for Vesicoureteral Reflux

30. Bladder Anomalies in Children

31. Exstrophy-Epispadias Complex

32. Prune-Belly Syndrome

33. Posterior Urethral Valves

34. Neuromuscular Dysfunction of the Lower Urinary Tract in Children

35. Functional Disorders of the Lower Urinary Tract in Children

36. Management of Defecation Disorders

37. Lower Urinary Tract Reconstruction in Children

28

# Clinical and Urodynamic Evaluation of Lower Urinary Tract Dysfunction in Children

*Duncan T. Wilcox, and Kyle O. Rove*

## Keywords

bowel and bladder dysfunction; lower urinary tract; pediatric urology; urodynamics; voiding dysfunction

## Questions

1. Under what conditions is a uroflow reading valid and interpretable?
    a. Postvoid residual (PVR) < 100 mL
    b. Voided volume > 150 mL
    c. Voided volume > 50 mL
    d. Electromyography (EMG) lag time < 6 seconds
    e. Any voided volume if there is no PVR
2. What are the five reference uroflow curve types?
    a. Stuttering, bell shaped, peaked, plateau, intermittent
    b. Staccato, intermittent, bell shaped, plateau, tower
    c. Tower, peaked, stuttering, normal, flat
    d. Plateau, intermittent, high, normal, staccato
    e. Continuous staccato, bell shaped, normal, inverted
3. EMG lag time less than 2 seconds corresponds to what potential patient presentation?

a. Normal voiding and bell-shaped uroflow curve
b. Overactive bladder and tower uroflow curve
c. Primary bladder neck dysfunction and plateau uroflow curve
d. Underactive bladder and staccato uroflow curve
e. Incontinence without PVR

4. What is a normal EMG lag time?
a. <2 seconds
b. 2 to 6 seconds
c. >6 seconds
d. 10 seconds
e. >12 seconds

5. EMG lag time greater than 6 seconds typically corresponds to what potential patient presentation?
a. Normal voider and bell-shaped uroflow curve
b. Overactive bladder and tower uroflow curve
c. Primary bladder neck dysfunction and plateau uroflow curve
d. Underactive bladder and staccato uroflow curve
e. Controlled (external sphincter) incontinence

6. Which are valid methods to define functional bladder capacity?
a. Voided volume on uroflow + PVR
b. (Age + 2) × 30 mL
c. Smallest voided volume on voiding diary
d. Largest voided volume on voiding diary
e. a + d

7. What is the significance of debris visualized within the bladder or collecting system on ultrasound?
a. This is a normal finding
b. Concern for ureterocele
c. Increased risk of symptomatic urinary tract infection
d. Increased risk of positive urine culture regardless of symptoms
e. Urothelial tumor

8. What test is recommended for all patients who present with lower urinary tract symptoms and no prior urologic diagnosis?
a. Urinalysis
b. Urodynamics
c. Serum creatinine

d. Uroflow
e. Renal bladder ultrasound

9. Which cutaneous finding over the sacral spine on exam is at highest risk for diagnosis of occult spinal dysraphism (tethered cord)?
    a. Cutaneous dimple less than 2.5 cm from anus
    b. Hemangioma
    c. Dermal sinus tract
    d. Symmetric bifid gluteal cleft
    e. hypertrichosis
10. What imaging modality is the gold standard for diagnosing spinal dysraphism?
    a. Computed tomography (CT) scan
    b. Ultrasound
    c. Positron emission tomography (PET) scan
    d. Magnetic resonance imaging (MRI)
    e. Videourodynamics
11. Under which conditions would it be appropriate to consider ultrasound of the spine to rule out spinal dysraphism?
    a. Newborn with open myelomeningocele
    b. 6-month-old with deviated gluteal cleft
    c. 3-year-old with gait problems and secondary enuresis
    d. Healthy 12-month-old with normal exam and cutaneous dimple 2 cm from the anal verge
    e. Family history of a tethered cord
12. What test/treatment is recommended in patients with lower urinary tract dysfunction who fail first-line, conservative therapies (e.g., behavioral modification, timed voiding, avoidance of bladder irritants)?
    a. 7-day bowel and bladder diary or 48-hour frequency-volume chart
    b. Formal videourodynamics
    c. Treatment with antimuscarinics
    d. Treatment with α-blocker
    e. Intradetrusor injection of botulinum toxin
13. Lower urinary tract dysfunction is associated with which of the following?

a. Constipation
b. Neuropsychiatric issues
c. Urinary tract infections
d. Vesicoureteral reflux
e. All of the above

14. What is an appropriate fill rate on cystometrogram for a pediatric patient?
    a. 5 mL/min
    b. 10 mL/min
    c. 15 mL/min
    d. 20% of expected bladder capacity per minute
    e. 5% to 10% of expected bladder capacity per minute
15. What finding(s) may be reduced on urodynamics when performed under sedation?
    a. Detrusor overactivity
    b. Compliance
    c. Detrusor leak point pressure
    d. Bladder capacity
    e. Internal sphincter control

## Answers

1. **c. Voided volume > 50 m**L. Uroflowmetry is affected by multiple parameters. However, the validity of a test is generally affected by voided volumes. For many studies of pediatric uroflowmetry, **50 mL is the cutoff for validity**. In adults, a higher value is used (>150 mL). Ideally, a voided volume greater than 50 mL and less than 115% of expected bladder capacity generates the best results. Volumes greater than 115% of expected capacity may also produce uroflow results that are spurious or misleading.
2. **b. Staccato, intermittent, bell shaped, plateau, tower**. The Standardization Committee of the International Children's Continence Society published a list of terminology standardization in the Journal of Urology in 2014. **This important document lists**

**common terms used to describe voiding dysfunction** and the requisite testing performed on these patients.

3. **b. Overactive bladder and tower uroflow curve**. EMG lag times of 2 to 6 seconds are considered normal. **Times less than 2 seconds are prevalent in patients with overactive bladder**. Times greater than 6 seconds indicate primary bladder neck dysfunction.
4. **b. 2 to 6 seconds**. EMG lag times of 2 to 6 seconds are considered normal and associated with a bell-shaped voiding curve. Times less than 2 seconds are prevalent in patients with overactive bladder. Times greater than 6 seconds indicate primary bladder neck dysfunction.
5. **c. Primary bladder neck dysfunction and plateau uroflow curve.** EMG lag times of 2 to 6 seconds are considered normal. Times less than 2 seconds are seen with bladder hyperactivity. **Times greater than 6 seconds indicate primary bladder neck dysfunction** associated with a uroflow curve that plateaus or flattens after an initial rise.
6. **e. a + d. Functional bladder capacity is a measure of day-to-day capacity of a patient's bladder.** This number should reflect how much a particular patient's bladder holds when full. This is reflected by both (a) voided volume on uroflow + PVR and (d) smallest voided volume on voiding diary. (b) corresponds to the patient's estimated bladder capacity (if older than 1 year). (d) does not reflect any clinically useful measure of the bladder.
7. **d. Increased risk of positive urine culture regardless of symptoms. Studies looking at the association of debris on ultrasound have shown increased association with positive culture**. These findings should be interpreted with caution in asymptomatic patients.
8. **a. Urinalysis. American Urological Association (AUA) guidelines for work-up of lower urinary tract symptoms recommend a urinalysis to rule out the presence of a urinary tract infection**. None of the other options are recommended for initial testing in new patients seen for lower urinary tract symptoms.

9. **b. Hemangioma. The risk of occult spinal dysraphism is different for each of the listed lesions**. From lowest risk to highest, they are: simple cutaneous dimple, symmetric bifid gluteal, hypertrichosis (hair tuft), dermal sinus tract, and hemangioma.
10. **d. Magnetic resonance imaging (MRI). MRI is the most sensitive and specific test to rule out any form of spinal dysraphism**. Spinal ultrasound can be used for neonates up to age 6 months in whom there is suspicion for dysraphic conditions, but this imaging modality has lower sensitivity. There is no role for CT or PET scans or videourodynamics for the diagnosis of possible dysraphism.
11. **b. 6-month-old with deviated gluteal cleft**. There is debate between use of ultrasound and MRI to rule out spinal dysraphism. MRI is more sensitive but more expensive than ultrasound. **For skin findings associated with spina bifida, lesions with low likelihood of diagnosis of spina bifida (like deviated gluteal cleft) may be more appropriate to use ultrasound if testing is being considered**. Inconclusive or positive ultrasounds can then be followed up with an MRI to confirm the diagnosis.
12. **a. 7-day bowel and bladder diary or 48-hour frequency-volume chart. Elimination diaries can pinpoint a number of issues that may not be elicited during a clinical encounter, including low functional capacity, infrequent voiding, nocturia, and polydipsia**. This should be used prior to more invasive evaluation or treatment.
13. **e. All of the above**. Constipation, neuropsychiatric issues, urinary tract infections, and vesicoureteral reflux are commonly associated with lower urinary tract dysfunction.
14. **e. 5% to 10% of expected bladder capacity per minute**. Pediatric patients have varied bladder capacities, so fill rates relative to expected bladder capacity should be considered. As such, absolute fill rates (e.g., 20 mL/min) might be too fast for a neonate with a bladder capacity of 80 mL. The recommended fill rate is 5% to 10% of expected capacity per minute.
15. a. **Destrusor overactivity**. Studies have demonstrated that the only parameter reliably affected by sedation or anesthesia is detrusor

overactivity. That is, if it were present normally, it would not be present on a sedated urodynamics study. Compliance, detrusor leak point pressure (dLPP), and bladder capacity are not generally affected.

## Chapter Review

1. History and physical should be performed to ascertain whether the patient has primarily a storage or emptying disorder. Further workup and treatment are then tailored as needed.
2. The important components of a complete history should include symptomatology, voiding frequency, bowel function, prior urinary tract infections, relevant family history, maternal prenatal history including screening ultrasonography, birth history, developmental milestones including toilet training, neuropsychiatric conditions, past medical and surgical history, social history, diet, and review of systems.
3. Seven-day bowel and bladder diaries and 48-hour frequency volume charts are invaluable in diagnosing simple and complex lower urinary tract voiding dysfunction.
4. Validated questionnaires can be used to assess symptoms (dysfunctional voiding score system [DVSS], dysfunctional voiding and incontinence score system [DVISS], and incontinence symptom index-pediatric [ISI-P]), quality of life (Pediatric Incontinence Questionnaire [PinQ]), and behavioral changes in response to underlying urinary dysfunction (Child Behavior Checklist [CBCL] and short screening instrument for psychological problems in enuresis [SSIPPE]).
5. Particular attention should be paid to the lower back on physical examination with both visual inspection (examine for dimples, deviations of the gluteal cleft, hair tufts, dermal sinuses and vascular malformations) and palpation (presence of sacrum, rule out mass).
6. Children presenting with lower urinary tract symptoms should have urinalysis testing, with important test details including the specific

gravity, protein or glucose, and the presence of white, red blood cells, or bacteria.

7. Isolated malodorous urine in the general population lacks sensitivity and specificity to detect positive urine culture in an otherwise asymptomatic patient.
8. Blood testing is not recommended unless there are specific concerns for concomitant upper tract disease.
9. Noninvasive urodynamics includes modalities like uroflowmetry, patch electromyography, and pelvic ultrasound.
10. Uroflowmetry provides useful information regarding flow characteristics during urination, including shape or pattern. Qmax is the most relevant quantitative variable when assessing bladder outflow.
11. Ultrasonography is another useful adjunct, noninvasive exam that can provide a number of different clinically important variables, including prevoid and postvoid bladder volumes to calculated a PVR, bladder wall thickness, rectal diameter, and presence of debris.
12. EMG provides for assessing coordination between the bladder and external sphincter to determine whether voiding is synergistic or dyssynergic and can also be used in conjunction with uroflowmetry to determine the lag time.
13. Urodynamics is the “gold standard” for evaluation and diagnosis of lower urinary tract dysfunction.
14. Universal strategies of antibiotic prophylaxis or urine cultures prior to urodynamics do not appear to be effective at preventing postprocedure urinary tract infection, which occurs 1.4% of the time.
15. Special considerations in pediatric urodynamics include use of sedation for catheter placement and/or the procedure itself and bladder fill rate, accounting for the varied expected bladder capacities across a wide range of patient ages and sizes.

29

# Management Strategies for Vesicoureteral Reflux

*Antoine E. Khoury, and Elias Wehbi*

## Keywords

bowel and bladder dysfunction; lower urinary tract; pediatric urology; urodynamics; voiding dysfunction

## Questions

1. The estimated prevalence of vesicoureteral reflux (VUR) in children with a urinary tract infection (UTI) is:
    a. 1%.
    b. 3%.
    c. 5%.
    d. 10%.
    e. 30%.
2. Which of the following statements regarding reflux is FALSE?
    a. Antenatally detected reflux is associated with a male preponderance.
    b. Antenatally detected reflux is usually low grade in boys as compared with that in girls.
    c. Antenatally detected reflux is usually bilateral in boys as compared with that in girls.
    d. When reflux is detected antenatally, renal impairment is frequently present at birth and is likely due to congenital

dysplasia.

e. The majority of reflux detected later in life occurs in females.

3. Which of the following statements regarding VUR in regard to the patient's race is TRUE?
    a. The incidence of VUR is equal in children of all races.
    b. The disparity in the incidence of VUR with respect to race becomes clearer in adulthood.
    c. The frequency of detected VUR is lower in female children of African descent.
    d. African and white infants have a similar incidence of reflux diagnosed on the basis of antenatal hydronephrosis.
    e. There is a clear understanding regarding the predisposition to reflux because many of the studies have included patients from different countries around the world.
4. Which of the following statements is FALSE in regard to the diagnosis and treatment of sibling VUR?
    a. On the basis of clinical judgment and the presence or absence of urinary tract infection (UTI), the patient's age should be taken into account in regard to the decision to proceed with intervention to diagnose sibling reflux.
    b. It is reasonable to prescribe antibiotic prophylaxis while the decision to diagnose sibling reflux is being made.
    c. Once sibling reflux has been diagnosed, the indications for correction differ from those for treating reflux in the general pediatric population diagnosed after UTI.
    d. Siblings who are younger than 5 years with normal imaging studies of the kidneys can be managed on the basis of clinical judgment, and it is not absolutely necessary to obtain a voiding cystogram.
    e. Siblings younger than 5 years who present with cortical renal defects have the most to lose by febrile UTIs in the presence of VUR.
5. Primary reflux is a congenital anomaly of the ureterovesical junction with a deficiency of the
    a. longitudinal muscle of the extravesical ureter, which results in an inadequate valvular mechanism.

b. longitudinal muscle of the intravesical ureter, which results in an inadequate valvular mechanism.
c. circumferential muscle of the extravesical ureter, which results in an inadequate valvular mechanism.
d. circumferential muscle of the intravesical ureter, which results in an inadequate valvular mechanism.
e. longitudinal and circumferential muscles of the intravesical ureter, which results in an inadequate valvular mechanism.

6. What is the accepted ratio of tunnel length to ureteral diameter found in most normal children without reflux?
   a. 5:1
   b. 4:1
   c. 3:1
   d. 2:1
   e. 1:1
7. Which of the following statements is TRUE regarding children with nonneurogenic neurogenic bladders?
   a. Constriction of the urinary sphincter occurs during voiding in a voluntary form of detrusor-sphincter dyssynergia.
   b. Gradual bladder decompensation and myogenic failure result from incomplete emptying.
   c. Gradual bladder decompensation and myogenic failure result from increasing amounts of residual urine.
   d. All of the above.
   e. None of the above.
8. Which of the following statements is TRUE in regard to secondary VUR?
   a. The most common cause of anatomic bladder obstruction in the pediatric population is posterior urethral valves, and VUR is present in a great majority of these children.
   b. Anatomic obstruction of the bladder is a common cause of secondary VUR in female patients.
   c. Patients with neurofunctional etiology for secondary VUR benefit from immediate surgical intervention to try to correct VUR.

d. A sacral dimple or hairy patch on the lower back is not a significant finding in regard to the evaluation and treatment of VUR.
e. The most common structural obstruction in male and female patients is the presence of a ureterocele at the bladder neck.

9. The complex anatomic relationships required of the ureterovesical junction may gradually be damaged by
    a. a decrease in bladder wall compliance.
    b. detrusor decompensation.
    c. incomplete emptying.
    d. subureteric injection of a bulking agent.
    e. a, b, and c.
10. What does the initial management of functional causes of reflux involve?
    a. Transurethral cystoscopic treatment
    b. Medical treatment
    c. Observation only
    d. Laparoscopic repair
    e. Videourodynamics
11. Signs or symptoms of bladder dysfunction include
    a. dribbling.
    b. urgency.
    c. incontinence.
    d. “curtsying” behavior in girls.
    e. all of the above.
12. Treatment of bladder dysfunction and detrusor overactivity regardless of its severity or cause is directed at:
    a. dampening overactive detrusor contractions.
    b. dilating the urethral sphincter.
    c. lowering intravesical pressures.
    d. a and c.
    e. all of the above.
13. There is a strong association between the presence of reflux in patients with neuropathic bladders and intravesical pressures of greater than:

a. 10 cm $H_2O$.
b. 20 cm $H_2O$.
c. 40 cm $H_2O$.
d. 60 cm $H_2O$.
e. 80 cm $H_2O$.

14. Bladder infections and their accompanying inflammation can also cause reflux by:
    a. lessening compliance.
    b. elevating intravesical pressures.
    c. distorting and weakening the ureterovesical junction.
    d. all of the above.
    e. none of the above.
15. Which system provides the current standard for grading reflux on the basis of the appearance of contrast in the ureter and upper collecting system during voiding cystourethrography?
    a. The Heikel and Parkkulainen system
    b. The International Classification system
    c. The Dwoskin and Perlmutter system
    d. The National Classification System
    e. The Dwoskin and Parkkulainen system
16. Which of the following is TRUE regarding accurately grading reflux with coexistent obstruction of the ipsilateral ureteropelvic junction (UPJ)?
    a. It is not possible.
    b. It is facilitated by obtaining a mercaptoacetyltriglycine (MAG3) lasix renal scan.
    c. It is facilitated by obtaining a renal bladder ultrasound.
    d. It is facilitated by obtaining a magnetic resonance urogram.
    e. It is facilitated by obtaining a radionuclide cystogram.
17. Which of the following is TRUE regarding the presence of fever?
    a. It represents eventual renal scarring.
    b. It correlates with greater than 100,000 bacterial colony-forming units.
    c. It may not always be a reliable sign of upper urinary tract involvement.

d. It increases the likelihood of discovering VUR.
e. c and d.

18. Complete evaluation including a voiding cystourethrogram (VCUG) and ultrasound are required for which of the following patients?
    a. An uncircumcised male infant with a febrile illness and a positive urine culture obtained through a bagged specimen
    b. A 3-year-old girl admitted to the hospital with pneumonia and found to have *Escherichia coli* on a urine culture without pyuria detected by microscopic analysis
    c. A female patient with recurrent culture and urinalysis proven to have afebrile UTIs and later found to have scarring on a dimercaptosuccinic acid (DMSA) scan
    d. Any child older than 5 years with documented UTIs
    e. None of the above
19. Which of the following is required for screening of older girls who present with asymptomatic bacteriuria?
    a. Renal bladder ultrasound
    b. VCUG
    c. VUDS
    d. DMSA renal scan
    e. No screening studies are required
20. Which of the following is TRUE regarding cystography?
    a. Cystography performed with a Foley catheter or while the patient is under anesthesia produces static studies that inaccurately screen for reflux or sometimes exaggerate its degree because of bladder overfilling.
    b. Cystography performed in the presence of excessive hydration may mask low grades of reflux because diuresis can blunt the retrograde flow of urine.
    c. Cystography may show reflux only during active infection when cystitis weakens the ureterovesical junction with edema or by increasing intravesical pressures.
    d. Cystography obtained during active infection will underestimate the grade of reflux because the endotoxins

produced by some gram-negative organisms can paralyze ureteral smooth muscle.

e. a, b, and c.

21. Radionuclide cystography:
    a. provides similar anatomic detail to that obtained with fluoroscopic cystography.
    b. is an accurate method for detecting and following reflux.
    c. is associated with more radiation exposure than is fluoroscopic cystography.
    d. is a less sensitive test than fluoroscopic cystography.
    e. provides more anatomic detail than fluoroscopic cystography.
22. Ultrasonography of the kidneys and bladder:
    a. is the diagnostic study of choice to initially evaluate the upper urinary tracts of patients with suspected or proven VUR.
    b. can effectively rule out reflux.
    c. should be performed every 2 to 3 years in patients with reflux who are medically managed.
    d. is the study of choice for assessing renal function.
    e. showing intermittent dilatation of the renal pelvis or ureter confirms the presence of reflux.
23. What is the best study for the detection of pyelonephritis and cortical renal scarring?
    a. Diethylenetriaminepentaacetic acid (DTPA) renal scan
    b. DMSA renal scan
    c. MAG3 renal scan
    d. computed tomography (CT) urogram
    e. Renal ultrasound
24. Which of the following is TRUE regarding urodynamic studies?
    a. They may be indicated in any child suspected of having a secondary cause for reflux (valves, neurogenic bladder, nonneurogenic neurogenic bladder, voiding dysfunction).
    b. They should be performed without the use of prophylactic antibiotics in children with secondary reflux.
    c. They help to direct therapy in patients with secondary reflux.
    d. They are more accurate when performed under sedation.
    e. a and c.

25. Which of the following is TRUE in regard to the evaluation of VUR?
    a. Routine cystoscopy is indicated in the workup of patients with VUR.
    b. The radiation doses with modern digital techniques have improved the anatomic detail, but the radiation dose with VCUG remains significantly higher than that of a radionuclide cystogram.
    c. Grading of reflux by VCUG and radionuclide cystography is similar and comparable between the two imaging modalities.
    d. Ultrasonography provides an alternative means of evaluating the presence or absence of VUR.
    e. Uroflowmetry is a valuable tool in the workup of a patient with VUR.
26. During ureteral development, a ureteral bud that:
    a. is medially (caudally) positioned from a normal takeoff at the trigone offers an embryologic explanation for primary reflux.
    b. is laterally (cranially) positioned from a normal takeoff at the trigone offers an embryologic explanation for primary reflux.
    c. fails to meet with the renal blastema offers an embryologic explanation for primary reflux.
    d. is laterally (cranially) positioned is often obstructed.
    e. fails to meet with the renal blastema is often obstructed.
27. In regard to the diagnosis of renal scars based on renal scintigraphy, which of the following is TRUE?
    a. An area of photopenia detected during an acute episode of pyelonephritis always represents renal scar.
    b. Photopenic areas may result from postinfection renal scarring and some renal dysplasia.
    c. Ultrasound is a sensitive and accurate diagnostic modality for renal scarring.
    d. Areas for photopenia detected during an acute episode of pyelonephritis that later resolve on a subsequent renal scan represent resolution of renal scarring.
    e. All of the above.
28. Which of the following is TRUE regarding hypertension?

a. In children and young adults, it is most commonly caused by reflux nephropathy.
b. It is not related to the grade of reflux or severity of scarring.
c. It is not associated with abnormalities of $Na^+$,$K^+$-ATPase activity.
d. All of the above.
e. None of the above.

29. Which of the following factors might contribute to the effects of reflux on renal growth?
    a. The congenital dysmorphism often associated with but not caused by reflux
    b. The number and type of urinary infections and their resultant nephropathy
    c. The quality of the contralateral kidney and its implications for compensatory hypertrophy
    d. The grade of reflux in the affected kidney
    e. All of the above
30. Which of the following statements is FALSE in regard to bladder and bowel dysfunction (BBD) and VUR?
    a. BBD lowers VUR resolution rates.
    b. BBD is associated with higher recurrence rates of VUR after successful endoscopic correction.
    c. BBD reduces the success rate of endoscopic implantation of dextranomer/hyaluronic copolymer (Dx/HA) and open surgical correction of VUR.
    d. BBD is associated with higher breakthrough infection rates.
    e. BBD is associated with an increased incidence of UTI after surgery.
31. The anatomy of patients with ureteral duplication typically follows the Weigert-Meyer rule, in which the upper pole ureter enters the bladder:
    a. distally and medially, and the lower pole ureter enters the bladder proximally and laterally.
    b. proximally and medially, and the lower pole ureter enters the bladder distally and laterally.

c. distally and laterally, and the lower pole ureter enters the bladder proximally and medially.
d. proximally and laterally, and the lower pole ureter enters the bladder distally and medially.
e. superior to the lower pole ureter.

32. Which of the following is not found to be associated with higher success rate of endoscopic correction of VUR?
    a. Volume of bulking agent used
    b. Surgeon experience
    c. Volcano-shaped mound with no hydrodistention
    d. Negative intraoperative cystography
    e. Utilization of the double hydrodistention-implantation technique
33. Which of the following accurately describes the state of the bladder during pregnancy?
    a. Urine volume decreases in the upper collecting system while the physiologic dilatation of pregnancy evolves.
    b. Bladder tone increases because of edema and hyperemia.
    c. Bladder changes predispose the patient to bacteriuria.
    d. All of the above.
    e. None of the above.
34. During pregnancy, the presence of VUR in a system already prone to bacteriuria may lead to increased morbidity. What is an additional risk factor?
    a. Renal scarring
    b. Tendency toward urinary infections
    c. Hypertension
    d. Renal insufficiency
    e. All of the above
35. Which of the following statements is considered to be FALSE regarding reflux management?
    a. Spontaneous resolution of VUR is common.
    b. High-grade VUR has a higher resolution rate when diagnosed in the newborn as compared with diagnosis in childhood.
    c. Reflux of sterile urine is a benign process that does not lead to significant renal damage.

d. The 2014 *New England Journal of Medicine* Randomized Intervention for Children with Vesicoureteral Reflux (RIVUR) study reported a 0.5% incidence of adverse reaction to prophylactic antibiotics and a 2% incidence of adverse reaction to the placebo.
e. Reflux cannot be accurately graded in patients with posterior urethral valves.

36. Common to each type of open surgical repair for reflux is the creation of:
    a. a valvular mechanism that enables ureteral compression with bladder filling and contraction.
    b. a mucosal tunnel for reimplantation having adequate muscular backing.
    c. a tunnel length of three times the ureteral diameter.
    d. a ureteral orifice aimed toward the trigone/bladder neck.
    e. all of the above.
37. Complete ureteral duplications with reflux can be best managed surgically by:
    a. separating the ureters and reimplanting them separately.
    b. a common sheath repair in which both ureters are mobilized with one mucosal cuff.
    c. performing an upper to lower ureteroureterostomy and reimplanting the lower ureter.
    d. performing a lower to upper ureteroureterostomy and reimplanting the upper ureter.
    e. none of the above.
38. Early postoperative obstruction can occur after a ureteral reimplant due to:
    a. edema.
    b. subtrigonal bleeding.
    c. a twist or angulation of the ureter.
    d. blood clots.
    e. all of the above.
39. If early postoperative obstruction occurs after a ureteral reimplant, the next step is:
    a. immediate nephrostomy tube placement.

b. immediate placement of a ureteral stent.
c. initial observation and diversion for unabating symptoms.
d. placement of both a nephrostomy tube and a ureteral stent.
e. reoperation.

40. Which of the following is TRUE regarding persistent reflux after ureteral reimplantation?
    a. It may be due to unrecognized secondary causes of reflux such as neuropathic bladder and severe voiding dysfunction.
    b. It seldom results from a failure to provide adequate muscular backing for the ureter within its tunnel.
    c. It may be repaired surgically by using minor submucosal advancements.
    d. It should be repaired immediately with a subureteric bulking agent.
    e. None of the above.
41. Which of the following is TRUE regarding the treatment of VUR?
    a. Since the widespread acceptance of endoscopic treatment, the indications for surgical correction differ between the open endoscopic and laparoscopic approaches.
    b. Long-term follow-up data support the durability of endoscopic injection therapy.
    c. All injection materials provide a similar success rate and are just as easily injected under similar circumstances.
    d. The accuracy of the needle entry point during endoscopic injection as well as the needle placement are important components for the success of the surgical procedure.
    e. The learning curve for endoscopic injection is similar to the learning curve for open surgical reimplantation.
42. Which of the following is TRUE regarding the laparoscopic approach for ureteral reimplantation?
    a. The advantages of this approach versus open surgery include smaller incisions, less discomfort, and quicker convalescence.
    b. As with other laparoscopic procedures, experience is essential to the success of this approach.
    c. Costs may be increased because of lengthier surgery and the expense of disposable equipment.

d. It is associated with a success rate higher than that of open surgery.
e. a, b, and c.

43. After discontinuation of continuous antibiotic prophylaxis (CAP) in toilet-trained children, who is likely to develop recurrent UTI?
    a. Patients with higher grades of VUR
    b. Uncircumcised male children
    c. Children with BBD
    d. All of the above
    e. a and c only
44. Which patients were more likely to have febrile or symptomatic recurrences in the RIVUR trial?
    a. Children with grade III or IV reflux at baseline
    b. Patients presenting with febrile index infection
    c. The presence of BBD at baseline
    d. All of the above
    e. a and c only
45. The 2011 American Academy of Pediatrics guidelines for management of the initial UTI in febrile infants and children 2 to 24 months of age recommend obtaining:
    a. a renal and bladder ultrasound and a VCUG during the febrile episode.
    b. a renal and bladder ultrasound and a VCUG 3 weeks after the febrile episode has resolved.
    c. a DMSA renal scan and, if positive, a VCUG.
    d. a renal and bladder ultrasound after confirmation of UTI by a properly collected urine specimen for culture and analysis.
    e. waiting for the second infection before performing any radiological testing.

## Answers

1. **e. 30%.** A meta-analysis of studies of children undergoing cystography for various indications has indicated that the

prevalence of VUR is estimated to be 30% for children with UTIs and approximately 17% in children without infection.

2. **b. Antenatally detected reflux is usually low grade in boys as compared with that in girls.** Reflux detected due to workup of antennal hydronephrosis is usually high grade and bilateral in boys as compared with that in girls.
3. **c. The frequency of detected VUR is lower in female children of African descent. One of the clear differences that has been established with several studies is the relative 10-fold lower frequency of vesicoureteral reflux in female children of African descent.**
4. **c. Once sibling reflux has been diagnosed, the indications for correction differ from those for treating reflux in the general pediatric population diagnosed after UTI.** If undertaken, imaging of the kidneys first, as well as the patient's age and history of UTI, can offer counsel with regard to the approach to sibling reflux screening. However, in any sibling in whom reflux is diagnosed, the indications for treatment remain the same as for general reflux in the pediatric population.
5. **b. Longitudinal muscle of the intravesical ureter, which results in an inadequate valvular mechanism.** Primary reflux is a congenital anomaly of the ureterovesical junction in which a deficiency of the longitudinal muscle of the intravesical ureter results in an inadequate valvular mechanism.
6. **a. 5:1. In Paquin's novel study, a 5:1 tunnel length–ureteral diameter ratio was found in normal children without reflux.**
7. **d. All of the above.** On the far end of this spectrum are children with nonneurogenic neurogenic bladders. Here, constriction of the urinary sphincter occurs during voiding in a voluntary form of detrusor-sphincter dyssynergia. Gradual bladder decompensation and myogenic failure result from incomplete emptying and increasing amounts of residual urine.
8. **a. The most common cause of anatomic bladder obstruction in the pediatric population is posterior urethral valves, and VUR is present in the great majority of these children.** This diagnosis is obviously limited to male patients; consequently female patients

have a lower incidence of anatomic bladder obstruction. **The most common structural obstruction in female patients is the presence of a ureterocele that prolapses and obstructs the bladder neck. Between 48% and 70% of patients with posterior urethral valves have vesicoureteral reflux, and relief of obstruction appears to be responsible for resolution of the reflux in a good number of those patients.** The presence of a neurologic disorder should prompt the clinician to treat on the basis of the primary etiology as opposed to proceeding with immediate surgical correction. One important aspect of the physical examination in children who present with VUR is detection of potential occult spinal dysraphism, which includes a thorough physical examination looking for sacral dimples, hairy patches, or gluteal cleft abnormalities.

9. **e. a, b, and c.** A decrease in bladder wall compliance, detrusor decompensation, and incomplete emptying gradually damages the complex anatomic relationships required of the ureterovesical junction. Subureteric injection of various bulking materials has not been shown to compromise UVJ function.
10. **b. Medical treatment.** The initial management of functional causes of reflux is medical. It is imperative that clinicians inquire about and determine the voiding patterns of children with reflux.
11. **e. All of the above.** In addition to a careful physical examination, signs or symptoms of bladder dysfunction include dribbling, urgency, and incontinence. Girls often exhibit curtsying behavior and boys will squeeze the penis in an attempt to suppress bladder contractions.
12. **d. a and c only.** Treatment of bladder dysfunction and detrusor overactivity, regardless of its severity or cause, is directed at decreasing overactive detrusor contractions and lowering intravesical pressures. Urethral dilation does not have a role in the routine treatment of bladder dysfunction.
13. **c. 40 cm $H_2O$.** There is a strong association between intravesical pressures of greater than 40 cm $H_2O$ and the presence of reflux in

patients with myelodysplasia and neuropathic bladders, although upper tract damage can also be seen in lower bladder pressures.

14. **d. All of the above. Bladder infections (UTIs) and their accompanying inflammation can also cause reflux by lessening compliance, elevating intravesical pressures, and distorting and weakening the ureterovesical junction.**
15. **b. The International Classification System.** The Heikel and Parkkulainen system gained popularity in Europe a few years before the Dwoskin and Perlmutter system became widely accepted in the United States. The International Classification System, devised in 1981 by the International Reflux Study, represents a melding of the two. It provides the current standard for grading reflux on the basis of the appearance of contrast in the ureter and upper collecting system during voiding cystourethrography.
16. **a. It is not possible.** Accurate grading of reflux is impossible with coexistent ipsilateral obstruction.
17. **e. c and d.** The presence of fever may be an indicator of upper urinary tract involvement but is not always a reliable sign. However, if fever (and presumably pyelonephritis) is present, the likelihood of discovering VUR is significantly increased. Fever can be associated with less than 100,000 bacterial colonies. Fever and pyelonephritis does not always lead to renal scarring.
18. **c. A female patient with recurrent culture and urinalysis proven to have afebrile UTIs and later found to have scarring on a dimercaptosuccinic acid (DMSA) scan.** The presence of culture-proven UTIs in the setting of an abnormal renal scan should raise the question of VUR; it is reasonable to proceed with a VCUG and ultrasound in those patients. The other clinical scenarios include a patient without pyuria and a clear alternative source for her fever as well as an infant diagnosed with UTI with a specimen obtained through a bagged collection. In those children the diagnosis of UTI should be questioned before proceeding with evaluation through cystogram and renal ultrasonography. Patients older than 5 years should not undergo immediate VCUG just on the basis of the presence of a UTI.

19. **e. No screening studies are required.** Although a renal bladder ultrasound may be helpful in some cases, it is not required in all older children who present with asymptomatic bacteriuria or UTIs that manifest solely with lower tract symptoms (afebrile simple cystitis). Cystography is reserved for those with abnormal upper tracts or recalcitrant infections. The initial focus should be more on voiding and bowel habits and fluid intake.
20. **e. a, b, and c.** Excessive hydration may mask low grades of reflux because diuresis can blunt the retrograde flow of urine. Some reflux is demonstrated only during active infections when cystitis weakens the ureterovesical junction with edema or by increasing intravesical pressures. In addition, cystograms obtained during active infections can overestimate the grade of reflux because the endotoxins produced by some gram-negative organisms can paralyze ureteral smooth muscle and exaggerate ureteral dilatation.
21. **b. Is an accurate method for detecting and following reflux.** Nuclear cystography is the scintigraphic equivalent of conventional cystography. Although the technique does not provide the anatomic detail of fluoroscopic studies, it is an accurate method for detecting and following reflux.
22. **a. Is the diagnostic study of choice to initially evaluate the upper urinary tracts of patients with suspected or proven vesicoureteral reflux.** Ultrasonography is the diagnostic study of choice to initially evaluate the upper urinary tracts of patients with suspected or proven VUR. However, the appearance of the kidneys on ultrasound does not correlate with the absence or presence of reflux or with its grade.
23. **b. DMSA renal scan.** Renal scintigraphy with technetium-99m–labeled DMSA is the best study for detection of pyelonephritis and the cortical renal scarring that sometimes results.
24. **e. a and c.** Urodynamic studies may be indicated in any child suspected of having a secondary cause for reflux (e.g., valves, neurogenic bladder, nonneurogenic neurogenic bladder, voiding dysfunction), and they help to direct therapy. Sedation or anesthesia may change bladder function.

25. **e. Uroflowmetry is a valuable tool in the workup of a patient with VUR.** Evaluation of the lower urinary tract cannot solely rely on imaging studies because reflux is considered to be a dynamic phenomenon. Uroflowmetry provides valuable information in the clinical assessment of these patients. **Modern management of reflux does not include routine evaluation through cystoscopy.** The radionuclide cystogram has historically been described as a technique that requires a significantly lower dose of radiation than a regular VCUG, but the advances with modern digital techniques have significantly narrowed the difference between these two imaging modalities. Unfortunately, ultrasound cannot reliably detect the presence or absence of VUR.
26. **b. Is laterally (cranially) positioned from a normal takeoff at the trigone offers an embryologic explanation for primary reflux.** As Mackie and Stevens have suggested, a ureteral bud that is laterally (cranially) positioned from a normal takeoff at the trigone offers an embryologic explanation for primary reflux, whereas those inferiorly (caudally) positioned are often obstructed.
27. **b. Photopenic areas may result from postinfection renal scarring and some renal dysplasia. Vesicoureteral reflux, particularly reflux of higher grades, may result in renal dysplasia, which often appears scintigraphically identical to postinfection pyelonephritic scars.** During an episode of active pyelonephritis, the renal scan may show an area of photopenia that later, if it persists, represents renal scarring secondary to the infection. Neither renal scan nor ultrasonography can accurately differentiate between renal dysplasia and renal scarring.
28. **a. In children and young adults it is most commonly caused by reflux nephropathy. Reflux nephropathy is the most common cause of severe hypertension in children and young adults, although the actual incidence is unknown.**
29. **e. All of the above.** Factors that might contribute to the effects of reflux on renal growth include the congenital dysmorphism often associated with reflux (30% of cases) but not caused by it; the number and type of urinary infections and their resultant nephropathy; the quality of the contralateral kidney and its

implications for compensatory hypertrophy; and the grade of reflux in the affected kidney.

30. **c. BBD reduces the success rate of endoscopic implantation of dextranomer/hyaluronic copolymer (Dx/HA) and open surgical correction of VUR.** However, BBD is associated with an increased incidence of UTI after surgery.
31. **a. Distally and medially, and the lower pole ureter enters the bladder proximally and laterally.** The anatomy of patients with ureteral duplication typically follows the Weigert-Meyer rule, wherein the upper pole ureter enters the bladder distally and medially and the lower pole ureter enters the bladder proximally and laterally.
32. **d. Negative intraoperative cystography.** Studies have not consistently shown increased outcome success with the volume of the bulking agent, surgeon experience, or techniques of volcano-shaped mound with no hydrodistention or the double hydrodistention-implantation technique.
33. **c. Bladder changes predispose the patient to bacteriuria.** Bladder tone decreases because of edema and hyperemia, which are changes that predispose the patient to bacteriuria. In addition, urine volume increases in the upper collecting system as the physiologic dilation of pregnancy evolves.
34. **e. All of the above.** It seems logical to assume that during pregnancy, the presence of VUR in a system already prone to bacteriuria would lead to increased morbidity. Maternal history also becomes a factor if past reflux, renal scarring, and a tendency to develop urinary infections are included. Women with hypertension and an element of renal failure are particularly at risk.
35. **d. The 2014 *New England Journal of Medicine* Randomized Intervention for Children with Vesicoureteral Reflux (RIVUR) study reported a 0.5% incidence of adverse reaction to prophylactic antibiotics and a 2% incidence of adverse reaction to the placebo.** Adverse reactions to antibiotics were reported in 2% of *both* the antibiotic prophylaxis and placebo groups.
36. **a. A valvular mechanism that enables ureteral compression with bladder filling and contraction.** Common to each technique

is the creation of a valvular mechanism that enables ureteral compression with bladder filling and contraction, thus reenacting normal anatomy and function. A successful ureteroneocystostomy provides a submucosal tunnel for reimplantation having sufficient length and adequate muscular backing. **A tunnel length of five times the ureteral diameter is cited as necessary for eliminating reflux.**

37. **b. A common sheath repair in which both ureters are mobilized with one mucosal cuff.** Approximately 10% of children undergoing antireflux surgery have an element of ureteral duplication. The most common configuration is a complete duplication that results in two separate orifices. This is best managed by preserving a cuff of bladder mucosa that encompasses both orifices. **Because the pair typically share blood supply along their adjoining wall, mobilization as one unit with a "common sheath" preserves vascularity and minimizes trauma.**
38. **e. All of the above.** Early after surgery, various degrees of obstruction can be expected of the reimplanted ureter. Edema, subtrigonal bleeding, and bladder spasms all possibly contribute. Mucus plugs and blood clots are other causes. Most postoperative obstructions are mild and asymptomatic and resolve spontaneously. More significant obstructions are usually symptomatic. Affected children typically present 1 to 2 weeks after surgery with acute abdominal pain, nausea, and vomiting.
39. **c. Initial observation and diversion for unabating symptoms.** The large majority of perioperative obstructions subside spontaneously, but placement of a nephrostomy tube or ureteral stent sometimes becomes necessary for unabating symptoms.
40. **a. It may be due to unrecognized secondary causes of reflux such as neuropathic bladder and severe voiding dysfunction. Other than technical errors, failure to identify and treat secondary causes of reflux is a common cause of the reappearance of reflux. Foremost among these secondary causes are unrecognized neuropathic bladder and severe voiding dysfunction.**

41. **d. The accuracy of the needle entry point during endoscopic injection as well as the needle placement are important components for the success of the surgical procedure.** The learning curve for endoscopic injection is believed to be different from that of open surgical reimplantation, but studies have not been carried out comparing these two surgical approaches for correction of VUR. Treatment is currently based on the same indications, and these indications do not differ between the different types of intervention.
42. **e. a, b, and c.** The advantages of this approach versus open surgery include smaller incisions, less discomfort, brief hospitalizations (although many centers now perform open reimplants on an outpatient basis), and quicker convalescence. As with other laparoscopic procedures, a learning curve must be climbed and experience is essential to the success of this approach. Laparoscopic reimplantation requires a team with at least two surgeons; the repair is converted from an extraperitoneal to an intraperitoneal approach; operative time is longer than with open techniques (although with experience it is now becoming gradually shorter); and cost is increased because of lengthier surgery and the expense of disposable equipment.
43. **e. a and c only.** Uncircumcised male children older than 1 year do not appear to be at higher risk for development of recurrent UTI after discontinuation of CAP.
44. **d. All of the above.** Children with grade III or IV reflux at baseline and patients presenting with febrile index infection and the presence of BBD at baseline were considered event modifiers in the RIVUR trial and appeared to benefit from CAP.
45. **d. A renal and bladder ultrasound after confirmation of UTI by a properly collected urine specimen for culture and analysis.** A VCUG is recommended only if the renal and bladder ultrasound is abnormal or if the child develops a second infection.

## Chapter Review

1. Approximately one-third of the siblings of patients with reflux will have reflux.
2. Reflux that is inherited is thought to be due to an autosomal dominant pattern.
3. There is a frequent association of constipation and encopresis with reflux and UTIs.
4. If both the ureteropelvic junction (UPJ) and ureterovesical junction (UVJ) require operative repair, the UPJ should be repaired first.
5. There is an association of renal maldevelopment with high grades of reflux.
6. The cardinal renal anomalies associated with reflux are multicystic dysplastic kidney and renal agenesis.
7. Women with UTIs and reflux who have undergone reimplantation will still be at significant risk for UTIs during pregnancy and should be monitored.
8. Almost 80% of low-grade and half of grade III reflux will resolve spontaneously.
9. Sterile reflux is benign.
10. Cohen's cross-trigonal technique of ureteral reimplantation is particularly well suited for small bladders and thick-walled bladders.
11. There is a 10% to 15% incidence of contralateral reflux after unilateral reflux is repaired.
12. Prophylactic bilateral reimplantation for unilateral reflux is not indicated.
13. Reflux is unlikely to be of any clinical significance in the absence of infection in a patient with normal bladder function.
14. The greatest risk for postinfection renal scarring is in the first year of life.
15. Reflux associated with a paraureteral diverticulum resolves at a similar rate to primary reflux.
16. There is a 10-fold lower frequency of vesicoureteral reflux in female children of African descent.
17. The endoscopic repair of reflux is less invasive and less durable than the open surgical repair.
18. A 5:1 tunnel length–ureteral diameter ratio should be achieved in antireflux surgery for best results.

19. Bladder infections (UTIs) and their accompanying inflammation can also cause reflux by lessening compliance, elevating intravesical pressures, and distorting and weakening the ureterovesical junction.
20. Modern management of reflux does not include routine evaluation through cystoscopy.
21. Vesicoureteral reflux, particularly reflux of higher grades, may be associated with renal dysplasia, which often appears scintigraphically identical to postinfection pyelonephritic scars.
22. Reflux nephropathy is the most common cause of severe hypertension in children and young adults.
23. In a duplex system where one ureter refluxes and surgical reconstruction is indicated, both ureters should have a common sheath reimplantation because the paired ureters typically share blood supply along their adjoining wall, and mobilization as one unit with a “common sheath” preserves vascularity and minimizes trauma.
24. Failure to identify and treat secondary causes of reflux is a common cause of the reappearance of reflux following correction. Foremost among these secondary cssses are unrecognized neuropathic bladder and severe voiding dysfunction.

30

# Bladder Anomalies in Children

*Christopher C. Roth*

## Key words

bladder agenesis; bladder diverticula; bladder duplication; bladder; cystitis cystica; fibroepithelial polyphemorrhagic cystitis; megacystis; nephrogenic adenoma; urachus

## Questions

1. All are true regarding the embryologic development of the urinary system **EXCEPT** which?
    a. Between the fourth and sixth weeks of gestation, the urorectal septum divides the endodermal cloaca into a ventral urogenital sinus and a dorsal rectum.
    b. Around the fourth to fifth month of gestation, the allantoic duct and ventral cloaca involute as the bladder descends into the pelvis.
    c. The obliterated urachus becomes the median umbilical ligament and connects the apex of the bladder with the umbilicus.
    d. Definitive nephrons within the embryologic kidneys will begin urine production around the 12th week of gestation, and the first signs of lower urinary tract anomalies may become evident.

e. Unlike the case in males, the entire female urethra is derived from the pelvic part of the urogenital sinus.

2. Which of the following statements best describes visualization of the fetal bladder?
    a. Transabdominal imaging is more sensitive than transvaginal imaging for early detection of the fetal bladder.
    b. The fetal bladder typically empties every 15 to 20 minutes, and nonvisualization of the bladder necessitates prolonged inspection to make sure that a full bladder was not missed.
    c. At 10 weeks of gestation the bladder will be visualized in 25% of fetuses.
    d. At 13 weeks of gestation the bladder will be visualized in 50% of fetuses.
    e. During the first trimester the normal bladder diameter is 12 mm or less.
3. In prenatally detected bladder anomalies,
    a. the incidence of bladder agenesis is approximately 1 per 600,000 gestations.
    b. bladder agenesis is most common in males.
    c. bladder exstrophy is most often associated with nonvisualization of the bladder and oligohydramnios.
    d. most cases of bladder exstrophy are diagnosed prenatally.
    e. visualization of the bladder is expected to be normal in cases of bilateral renal dysplasia.
4. Regarding megacystis,
    a. the bladder is dilated secondary to high-grade reflux.
    b. this finding indicates that lower urinary tract obstruction (LUTO) is present.
    c. during the first trimester a normal bladder should be 6 mm or less.
    d. beyond the first trimester a bladder that fails to empty after 15 minutes is abnormal.
    e. this finding is always associated with a dilated posterior urethra.
5. Genetic evaluation of a fetus with megacystis is likely to demonstrate:

a. a definitive genetic diagnosis in most cases.
b. a decreased rate of expected genetic anomalies based on maternal age.
c. a decreased rate of expected genetic anomalies based on nuchal translucency.
d. trisomy 13 or 18.
e. none of the above.

6. Improved outcomes for megacystis have been noted when all are present **EXCEPT** which?
   a. Aneuploidy
   b. Initial bladder dilation of 12 mm or less
   c. Resolution of megacystis by 23 weeks of gestational age
   d. Normal amniotic fluid levels
   e. Isolated bladder findings without associated congenital anomalies
7. Megacystis microcolon intestinal hypoperistalsis (MMIH) syndrome:
   a. is best described as the most common cause of megacystis on routine surveillance.
   b. is most often detected prenatally by the gastrointestinal findings.
   c. has more morbidity for urologic than enteric sequelae.
   d. has no known survivors beyond the neonatal period.
   e. is suspected of being secondary to dysfunction of actin smooth muscle.
8. Bladder duplication will usually be associated with:
   a. duplex ureters and collecting systems.
   b. duplication of the urethra.
   c. less than 25% incidence of gastrointestinal (GI) anomalies.
   d. diurnal continence.
   e. genital anomalies.
9. Which statement is most accurate regarding congenital bladder diverticula?
   a. A majority are diagnosed prenatally.
   b. Most occur through a hiatus in the bladder wall and consist of mucosal lining only.

c. The reported incidence is 20% of the population.
d. A cystogram offers little information over ultrasound imaging.
e. Most diverticula will require surgical resection.

10. Diverticula secondary to connective tissue disorders:
    a. usually present as a single diverticulum.
    b. have walls composed of smooth muscle as well as mucosa.
    c. are best managed surgically before symptoms.
    d. have an unlikely chance of recurrence after resection.
    e. are often associated with increased incidence of urothelial tumors.
11. Regarding urachal anomalies,
    a. urachal squamous cell carcinoma has been reported in some children.
    b. most cases of a patent urachus are associated with bladder outlet obstruction.
    c. resection of a urachal cyst is not recommended after treatment of a prior infection.
    d. a patent omphalomesenteric duct can mimic a patent urachus.
    e. surgical management is best performed with minimally invasive surgery.
12. Which statement best describes noninflammatory bladder conditions?
    a. Bladder hemangiomas are often multiple.
    b. Bladder hemangiomas often require radical resection.
    c. Bladder hemangiomas are associated with Klippel-Trenaunay or Sturge-Weber syndrome.
    d. Fibroepithelial polyps are the most common cause of bladder masses in children.
    e. Fibroepithelial polyps are difficult to treat endocscopically.
13. Nephrogenic adenoma:
    a. is more common in children than in adults.
    b. often demonstrates malignant transformation.
    c. rarely recurs after local treatment.
    d. is best treated by a partial cystectomy.
    e. can be seen as a reaction to infection or urolithiasis.

14. When the diagnoses of cystitis cystica and eosinophilic cystitis is being considered, it is important to note that:
    a. cystitis cystica more commonly occurs in males.
    b. eosinophilic cystitis can be treated with steroids, antihistamines, and antibiotics.
    c. eosinophilic cystitis more commonly occurs in females.
    d. eosinophilic cystitis in young children will almost always progress.
    e. cystitis cystica treatment typically includes steroids.

## Answers

1. **d. Definitive nephrons within the embryologic kidneys will begin urine production around the 12th week of gestation, and the first signs of lower urinary tract anomalies may become evident.** Definitive nephrons within the embryologic kidneys will actually begin to produce urine prior to the 12th week of gestation. It is around the 10th week of gestation that the first signs of lower urinary tract anomalies may become evident.
2. **b. The fetal bladder typically empties every 15 to 20 minutes and nonvisualization of the bladder necessitates prolonged inspection to make sure that a full bladder was not missed**. The timing of ultrasound findings is important and due to the frequent cycling of the normal fetal bladder. Careful observation is needed prior to determining that a bladder is not visualized.
3. **a. The incidence of bladder agenesis is approximately 1 per 600,000 gestations.** Bladder agenesis incidence is approximately 1 per 600,000 and is more common in females. Bladder exstrophy is typically associated with normal amniotic fluid levels, and most cases are diagnosed postnatally.
4. **c. During the first trimester a normal bladder should be 6 mm or less.** Megacystis commonly refers to a finding on imaging and has a varied differential beyond dilation secondary to reflux. It may or may not be associated with bladder outlet obstruction. During the first trimester, the definition of megacystis is based on size,

with the normal bladder being 6 mm or less in size. Beyond the first trimester, megacystis is defined as a bladder that does not empty during 45 minutes of observation.

5. **d. Trisomy 13 or 18**. Genetic testing in patients with the finding of megacystis is not universal, and most cases will not have a genetic diagnosis. When genetic testing is performed, the most common anomalies are trisomy 13 and 18. Genetic anomalies in patients with megacystis exceed the expected rates when maternal age and nuchal translucency are controlled for.
6. **a. Aneuploidy**. When genetic testing is performed, positive findings are associated with poor outcomes. All other factors listed are predictors of improved outcomes.
7. **e. Is suspected of being secondary to dysfunction of actin smooth muscle**. MMIH is a rare cause of megacystis and is most often detected by the bladder findings. The urologic sequelae are significant, although morbidity is most often due to enteric complications. Although MMIH was once thought to be incompatible with life, survival beyond the neonatal period is now well documented.
8. **e. Genital anomalies**. Most cases of bladder duplication will be associated with genital anomalies and the urethra may or may not be duplicated. Over 25% will have GI anomalies, and continence is difficult to predict based on the anatomy alone.
9. **b. Most occur through a hiatus in the bladder wall and consist of mucosal lining only**. A cystogram is the definitive test for diagnosing bladder diverticula; most instances are diagnosed only following symptomatic presentation. The reported incidence is 1.7%, and most cases will not require surgical resection.
10. **b. Have walls composed of smooth muscle as well as mucosa**. Diverticula associated with connective tissue disorders will often present with multiple diverticula, and the typical histology will show smooth muscle and mucosa in the wall of the diverticulum. Surgical management can be chosen when symptomatic, and recurrence is to be expected.
11. **d. A patent omphalomesenteric duct can mimic a patent urachus**. Adenocarcinoma of urachal anomalies has been reported

in adults and is rare in children, but this is not true of squamous cell carcinoma. Most cases of a patent urachus are not associated with bladder outlet obstruction. Resection of a urachal cyst is recommended after treatment of a prior infection. Surgical management of urachal anomalies can be effectively performed with either open or minimally invasive surgery.

12. **c. Bladder hemangiomas are associated with Klippel-Trenaunay or Sturge-Weber syndrome**. Hemangiomas are often singular and can be associated with known syndromes. Fortunately, simple resection can be curative. Fibroepithelial polyps are rare causes of bladder masses and can usually be managed simply with endoscopic resection.
13. **e. Can be seen as a reaction to infection or urolithiasis.** This mostly benign lesion is usually described as a reaction to infection, lithiasis, or injury. Nephrogenic adenomas are more common in adults than in children. Recurrence after resection (usually cystoscopic rather than open) reaches 80%, with a latency of 4 years.
14. **b. Eosinophilic cystitis can be treated with steroids, antihistamines, and antibiotics.** Eosinophilic cystitis occurs more commonly in boys. When it occurs in young children, it will often be self-limited. Inflammatory cells with numerous eosinophils are seen throughout all layers of the biopsy and can be treated with a combination of steroids, antihistamines, and antibiotics. Cystitis cystica occurs more commonly in females and antibiotics are a component of treatment.

## Chapter Review

1. Persistent megacystis in the background of oligohydramnios is concerning for LUTO; megacystis that resolves prior to delivery may not have any postnatal sequelae.
2. MMIH syndrome may present with megacystis noted only prenatally.
3. Nondilated anomalies are the most severe forms of congenital urologic malformations, such as bladder exstrophy. Normal amniotic

fluid levels are usually found.

4. Bladder agenesis is compatible with life only if the ureters drain ectopically.
5. Children with connective tissue disorders can have multiple bladder diverticula that may have to be treated owing to stasis-related infections or stones.
6. Urachal anomalies are usually detected postnatally because of umbilical drainage.
7. Infected urachal remnants are treated with antibiotics, followed by surgical excision.
8. Urachal remnants can be confused for a persistent omphalomesenteric duct on preoperative workup.
9. Asymptomatic incidentally found urachal remnant excision should be discussed with family because of the potential for malignant transformation in adulthood.
10. Acquired bladder diverticula can become large enough to cause urinary obstruction or create stasis leading to recurrent urinary tract infections.
11. After resection of nephrogenic adenomas, routine cystoscopy is necessary because of an 80% recurrence rate in children.
12. Cystitis cystica can mimic a bladder tumor but is benign and is best treated with long-term antibiotic prophylaxis.
13. Eosinophilic cystitis has no “typical” appearance and requires biopsy for diagnosis.
14. Cyclophosphamide therapy can lead to hemorrhagic cystitis. For prevention, mesna is given simultaneously, and oral sodium pentosan polysulfate shows promise.